7 **(a)** Find $\int x^3\left(15x+\frac{11}{\sqrt[3]{x}}\right)\mathrm{d}x$. **[5]**

(b) Show that $\int_0^8 x^3\left(15x+\frac{11}{\sqrt[3]{x}}\right)\mathrm{d}x = a\times 2^{11}$, where a is a positive integer to be determined. **[3]**